

Unit 33: Exponential Change and Separable Differential Equations

1. The idea, in plain words

So far, every derivative problem has started with a function and asked you to find its rate of change. This unit flips that around: you're given the rate of change (as an equation involving $\frac{dy}{dx}$), and your job is to find the original function y that produces it. An equation like this — one that involves a derivative — is called a differential equation, and solving it means finding the family of functions y that satisfy it.

A “separable” differential equation is one where you can physically pull the x 's and y 's apart onto opposite sides of the equation. Picture $\frac{dy}{dx}$ as a fraction (it isn't technically one, but it behaves enough like one for this purpose) — if the right-hand side factors into “a piece with only x 's” times “a piece with only y 's,” you can shuffle every y term (along with dy) to one side, and every x term (along with dx) to the other side.

Once separated, you just integrate both sides. The left side gives you something in terms of y ; the right side gives you something in terms of x ; and you tie them together with a single constant of integration. If possible, you then solve algebraically for y to get an explicit solution — a genuine formula for y in terms of x . Sometimes solving for y isn't practical, and you leave the answer in implicit form instead (an equation relating x and y , without y isolated).

A famous special case worth deriving once and remembering forever: the Law of Exponential Change. Whenever a quantity's rate of change is proportional to the quantity itself ($\frac{dy}{dt} = ky$), separating and integrating always produces the same shape of answer: $y = y_0 e^{kt}$, where y_0 is the starting amount. This is exactly the formula behind population growth, radioactive decay, compound interest, and Newton's Law of Cooling — and now you'll see precisely why it always takes that shape, instead of just being handed the formula.

2. Toolbox

The separation-and-integration procedure:

1. Rewrite $\frac{dy}{dx} = f(x)g(y)$ by moving all y -terms (with dy) to one side, and all x -terms (with dx) to the other:

$$\frac{dy}{g(y)} = f(x) dx$$

2. Integrate both sides:

$$\int \frac{dy}{g(y)} = \int f(x) dx$$

3. Add a single constant of integration (the two separate $+C$'s from each side combine into one).
4. Solve for y explicitly if possible; if not, leave the relationship implicit.
5. If given an initial condition (like $y(0) = y_0$), plug it in to solve for the specific value of the constant.

The Law of Exponential Change (a direct consequence of separating $\frac{dy}{dt} = ky$):

$$\frac{dy}{dt} = ky \quad \implies \quad y = y_0 e^{kt}$$

where $y_0 = y(0)$ is the initial amount, and $k > 0$ means growth, $k < 0$ means decay.

Newton's Law of Cooling (a shifted version of the same idea): if $\frac{dT}{dt} = -k(T - T_s)$ (where T_s is the surrounding/ambient temperature), then

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

3. Common mistakes

- Trying to integrate before separating. You cannot integrate $\frac{dy}{dx} = f(x)g(y)$ directly — the variables must be fully separated onto opposite sides first.
- Forgetting the constant of integration, or adding two separate constants (one from each side) instead of combining them into a single overall constant.
- Losing track of signs when solving for y after integrating, especially when a $\ln|y|$ term needs to be exponentiated back out.
- Forgetting to apply the initial condition, if one is given — a general solution with an unresolved constant is not the final answer if you were handed a starting value.
- Trying to force a non-separable equation into this framework. Always double-check that the right-hand side genuinely factors into “ x -stuff times y -stuff” before attempting to separate.

4. Problem Set

Warm-up

Find the general solution to each differential equation.

1. $\frac{dy}{dx} = x$
2. $\frac{dy}{dx} = 2y$
3. $\frac{dy}{dx} = \frac{x}{y}$
4. $\frac{dy}{dx} = 3x^2$, with $y(0) = 5$
5. $\frac{dy}{dx} = \frac{y}{x}$ (for $x > 0$)
6. $\frac{dy}{dx} = -2y$, with $y(0) = 10$

Standard

7. $\frac{dy}{dx} = xe^{-y}$
8. $\frac{dy}{dx} = \frac{\cos x}{y}$
9. $\frac{dy}{dx} = y \cos x$
10. $\frac{dy}{dx} = x^2y$, with $y(0) = 2$
11. $\frac{dy}{dx} = 2xy^2$
12. $\frac{dy}{dx} = e^{x+y}$ (rewrite using exponent rules first!)

Challenge

13. Solve $\frac{dy}{dx} = \sin x \cdot e^{-y}$.
14. Starting from $\frac{dy}{dt} = ky$, separate and integrate to derive $y = Ce^{kt}$, then explain how the constant C relates to $y_0 = y(0)$.
15. Solve $\frac{dy}{dx} = y^2 \sin x$, with $y(0) = 1$.
16. Solve $\frac{dy}{dx} = \frac{x+1}{y-2}$, with $y(0) = 4$.
17. Solve $\frac{dy}{dx} = -xy$, with $y(0) = 3$.

Applied

18. A population grows according to $\frac{dP}{dt} = 0.04P$, with $P(0) = 1000$. Find $P(t)$, and find the population after 20 years.
19. An object cools according to Newton's Law of Cooling: $\frac{dT}{dt} = -0.1(T - 70)$, with $T(0) = 200$ (ambient temperature 70°). Find $T(t)$, and find the temperature after 10 minutes.
20. A radioactive sample decays according to $\frac{dA}{dt} = -0.05A$, with $A(0) = 200$ grams. Find $A(t)$, and find the time it takes to decay to 100 grams.
21. A drug's concentration in the bloodstream decreases according to $\frac{dC}{dt} = -0.3C$, with $C(0) = 80$ mg. Find $C(t)$, and find the concentration after 5 hours.